

BEE 208 — Question 3: Motor Starting Current Checker

Control Statement: if...else | Faculty of Engineering, ATU

1. ALGORITHM

Step 1: Start

Step 2: Declare variables ratedCurrent and startingCurrent (double)

Step 3: Prompt user to enter rated current

Step 4: Read ratedCurrent

Step 5: Prompt user to enter starting current

Step 6: Read startingCurrent

Step 7: If startingCurrent > 3 × ratedCurrent, display 'High starting current. Use proper motor starter.'

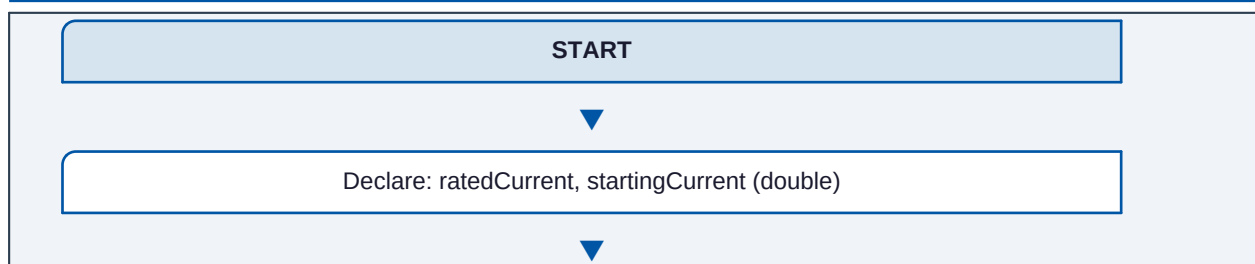
Step 8: Else display 'Starting current is acceptable.'

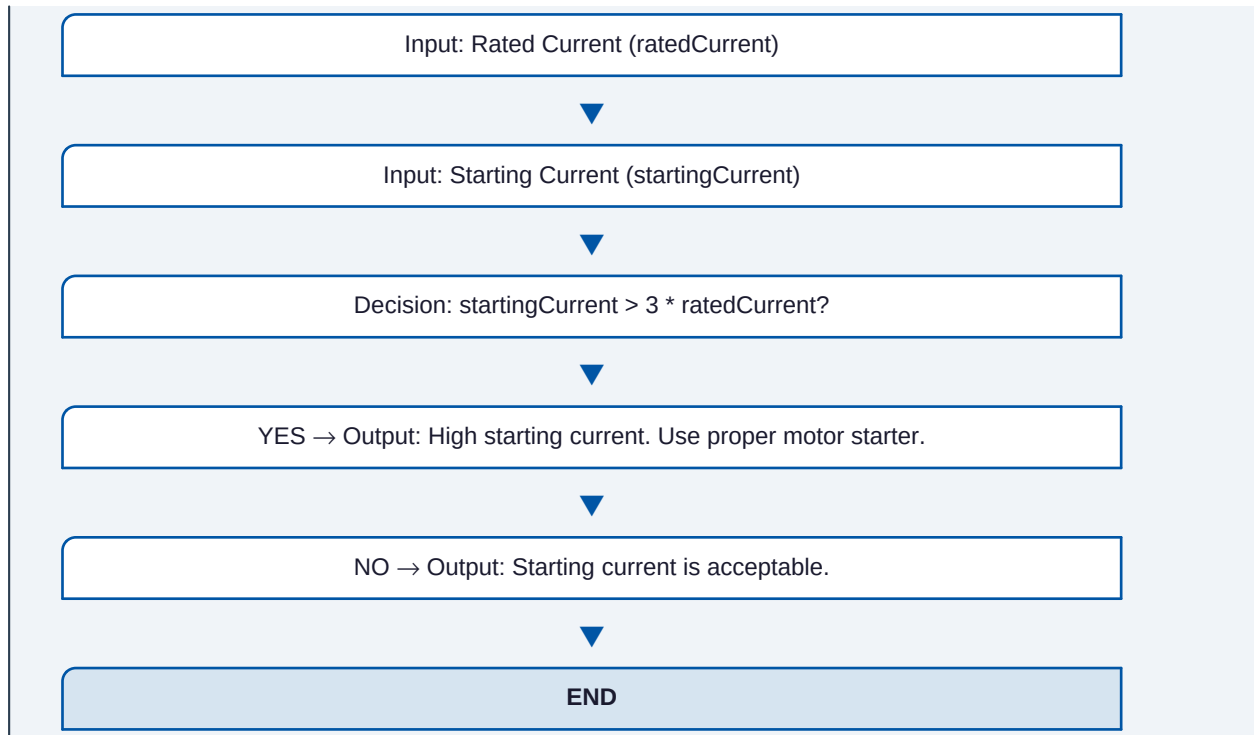
Step 9: End

2. PSEUDOCODE

```
BEGIN
DECLARE ratedCurrent, startingCurrent AS DOUBLE
OUTPUT 'Enter Rated Current (A): '
INPUT ratedCurrent
OUTPUT 'Enter Starting Current (A): '
INPUT startingCurrent
IF startingCurrent > 3 * ratedCurrent THEN
OUTPUT 'High starting current. Use proper motor starter.'
ELSE
OUTPUT 'Starting current is acceptable.'
END IF
END
```

3. FLOWCHART (Structured Diagram)





4. TEST RESULTS

Test Item	Value / Description	Test Item	Value / Description
Input	ratedCurrent=10A, startingCurrent=40A	Expected Output	High starting current. Use proper motor starter.
Input	ratedCurrent=10A, startingCurrent=25A	Expected Output	Starting current is acceptable.